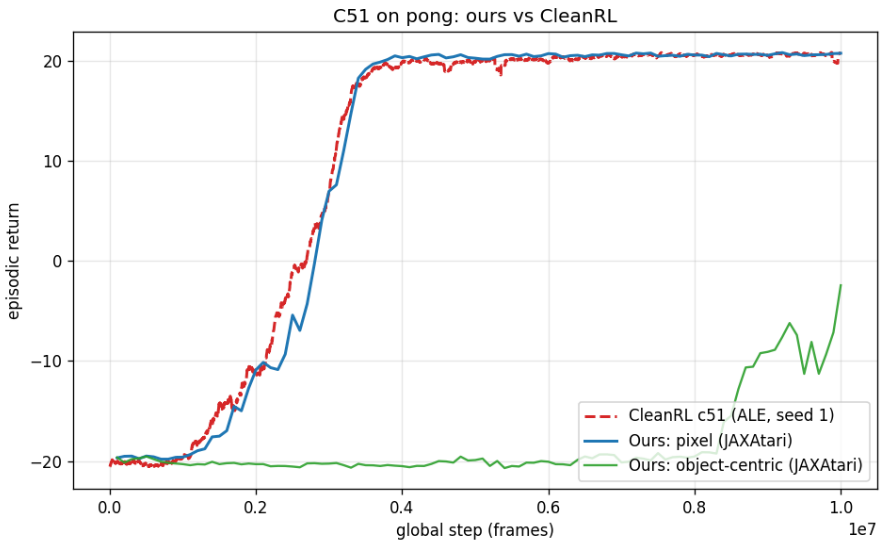
C51 (Categorical DQN) on Pong: progress report

Group 27 • Topic 27 (C51, IQN, Rainbow, IMPALA) • JAXAtari baselines

C51 is implemented in JAX/Flax as a single self-contained file supporting both observation modes (pixel with a CNN, object-centric with an MLP). **Both modes solve Pong.**

Mode	Network	Final return (10M)	Reference
Pixel	CNN	+**20.8**	CleanRL C51 ≈ +20
Object-centric	MLP	+**18.0**	no OC reference



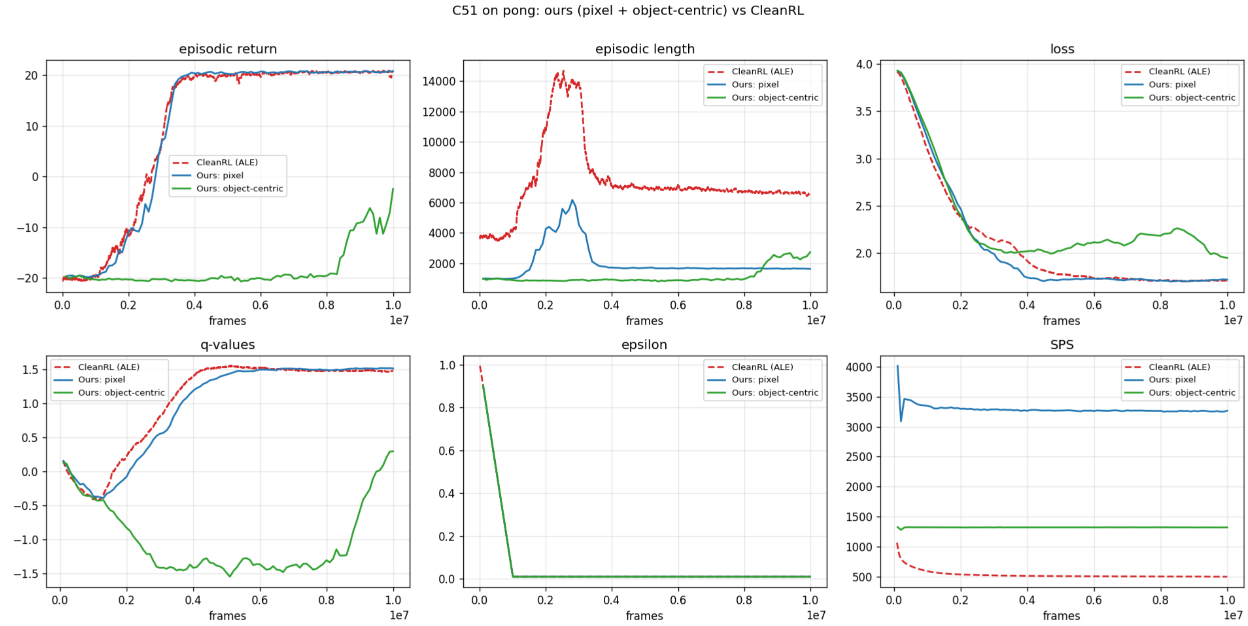
Episodic return on Pong (10M frames): our pixel and object-centric runs vs the CleanRL `c51_atari_jax` reference curve (ALE). Our pixel matches CleanRL; object-centric learns slower but reaches +18.

Object-centric finding. Object-centric observations needed normalization (the raw pixel-coordinate features were saturating the MLP); adding `NormalizeObservationWrapper` plus a 512×512 MLP fixed it. Same fix will apply to IQN, Rainbow and IMPALA.

Questions for feedback

1. Pixel matches the reference. Enough to call C51 pixel done, or do you want multi-seed variance bands?
2. OC uses `Normalize` $+512 \times 512$ instead of CleanRL's CartPole MLP. OK to keep?

Training metrics: ours vs CleanRL (10M)



Six metrics, three curves each: CleanRL `c51_atari_jax` (ALE), our pixel, and our object-centric. Loss, epsilon and q-values track CleanRL closely; SPS is higher for ours even at 8 envs (JAXAtari runs the env on GPU). Episode length is counted differently by JAXAtari than ALE.